$$P(\text{get a King}) = \frac{4}{52} = \frac{1}{13}$$

$$P(\text{get a King } \underline{\text{given}} \text{ it is a "face card"}) = \frac{4}{12} = \frac{1}{3}$$

"conditional probability"

We write

$$P(\text{get a King} \mid \text{it is a "face card"}) = \frac{1}{3}$$

Events : A, B, C

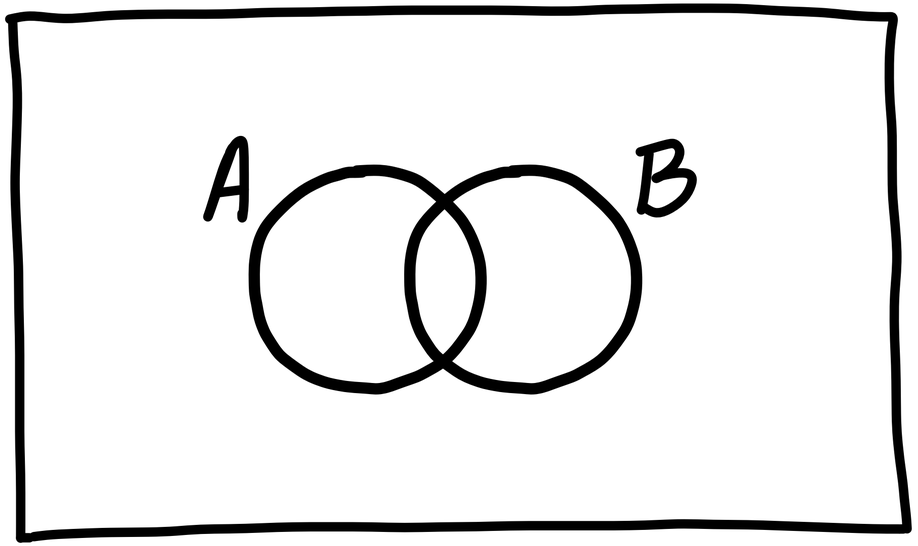
Let A = the event that we get a King

Let B = the event that we get a face card

Then

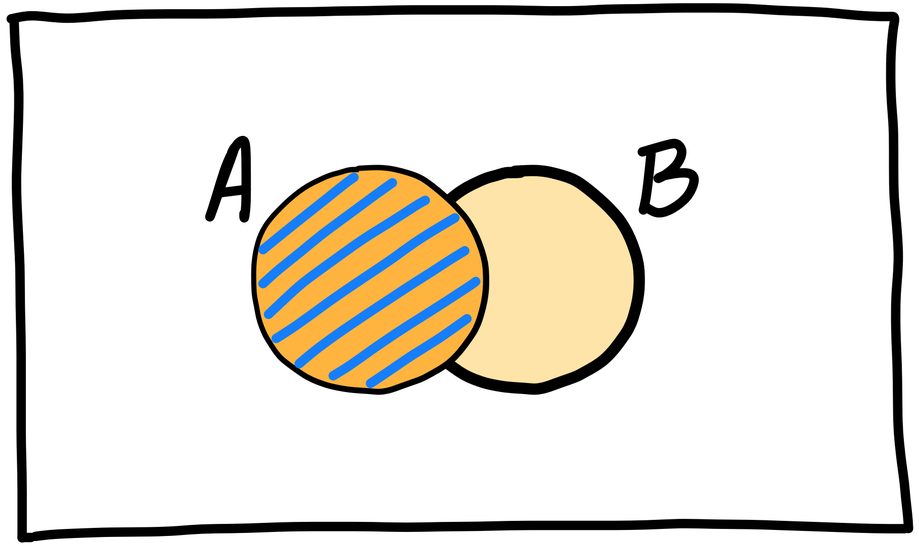
$$P(A|B) = \frac{1}{3}$$

Venn Diagrams: Probability as Area



Venn Diagrams: Probability as Area

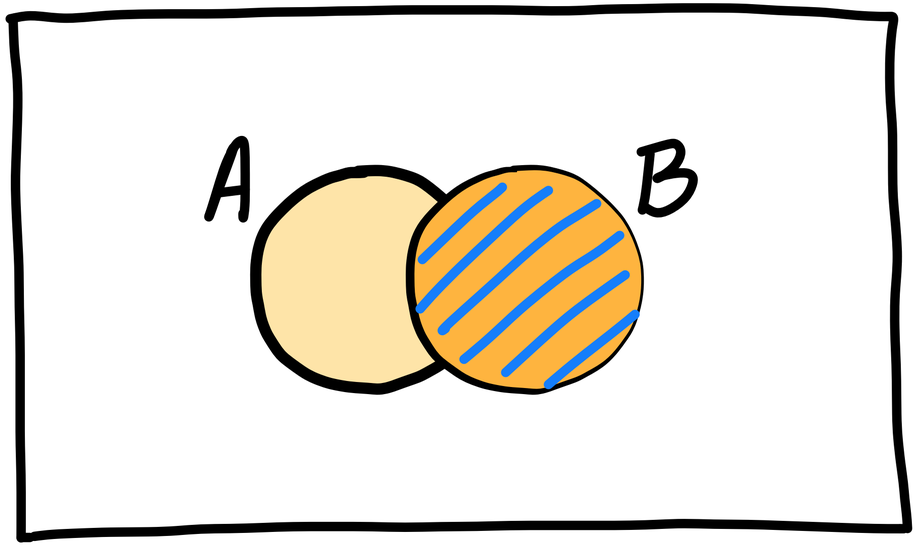
$A \cup B$



$P(A \cup B) = \boxed{P(A)} +$

Venn Diagrams: Probability as Area

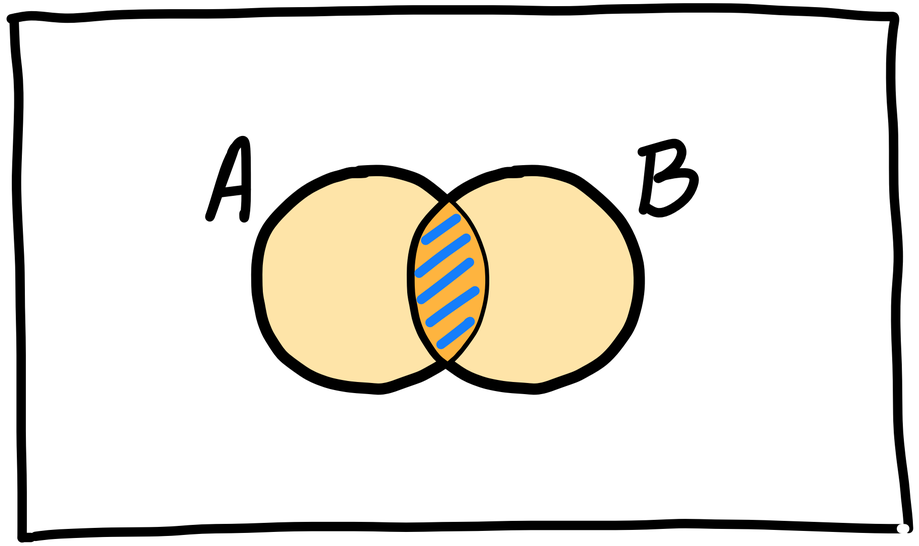
$A \cup B$



$$P(A \cup B) = P(A) + \boxed{P(B)}$$

Venn Diagrams: Probability as Area

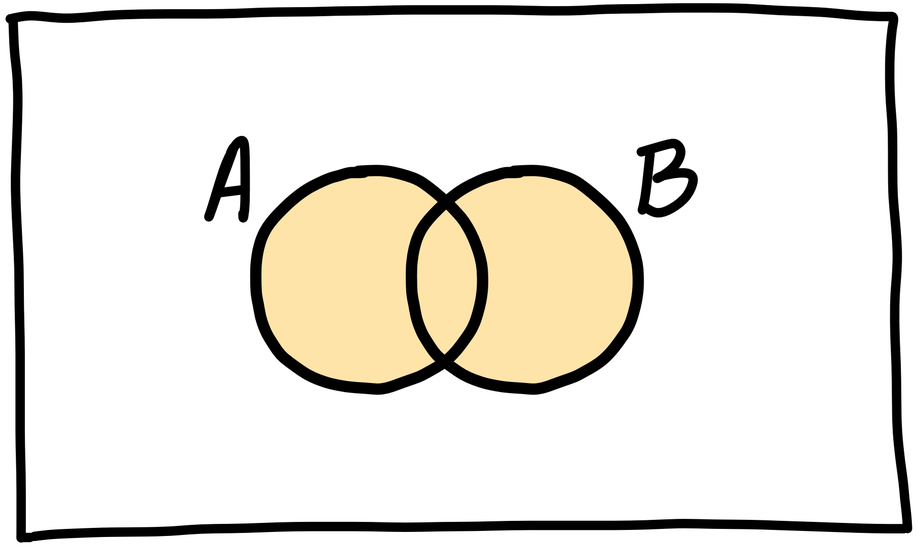
$A \cup B$



$$P(A \cup B) = P(A) + P(B) - \boxed{P(A \cap B)}$$

Venn Diagrams: Probability as Area

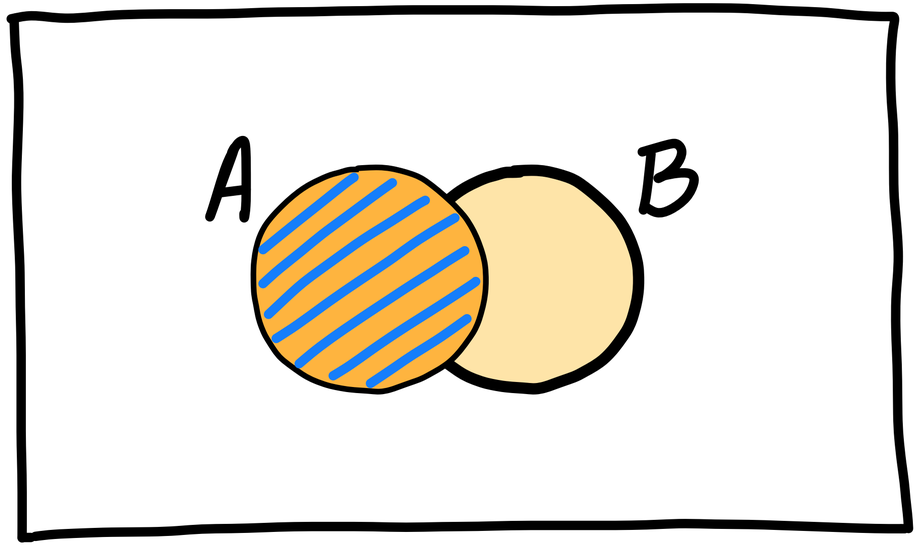
$A \cup B$



$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Venn Diagrams: Probability as Area

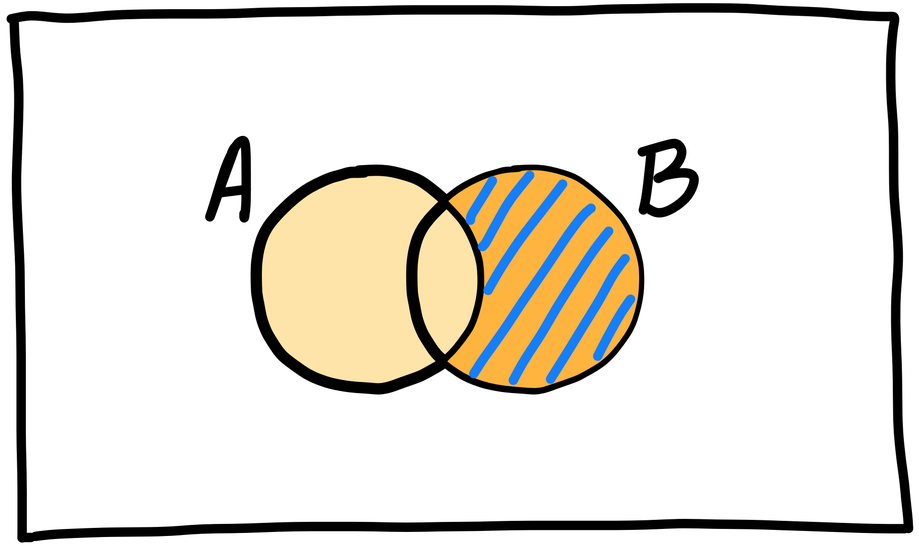
$A \cup B$



$P(A \cup B) = \boxed{P(A)} +$

Venn Diagrams: Probability as Area

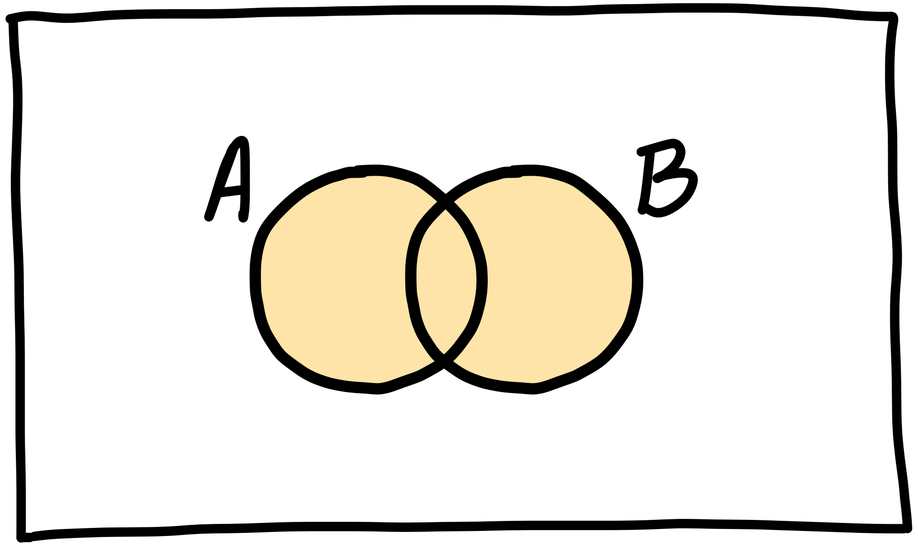
$A \cup B$



$$P(A \cup B) = P(A) + \boxed{P(B \cap A^c)}$$

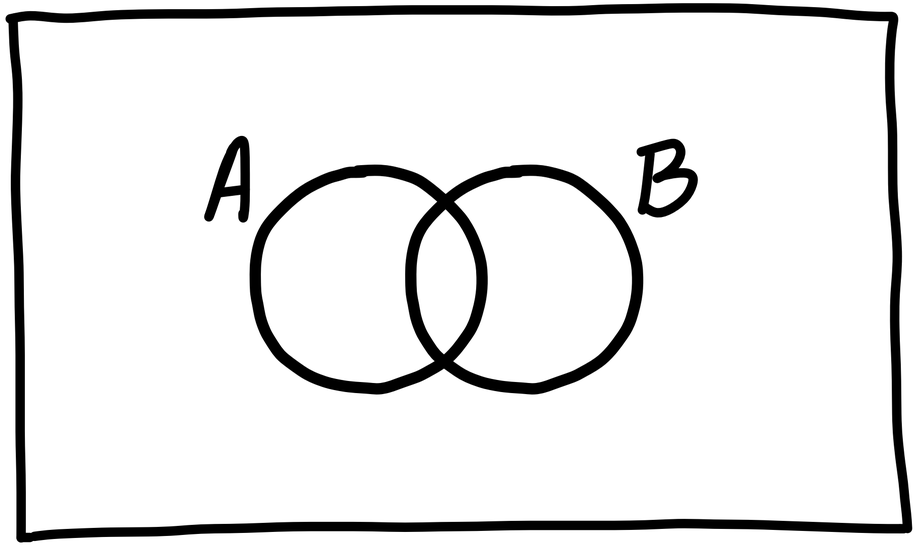
Venn Diagrams: Probability as Area

$A \cup B$



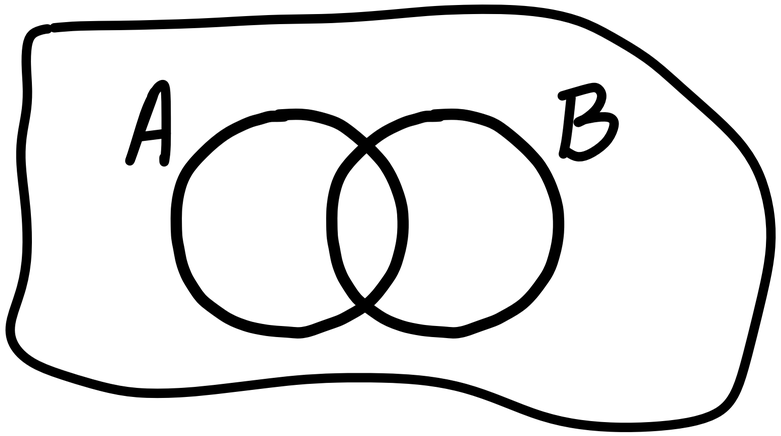
$$P(A \cup B) = P(A) + P(B \cap A^c)$$

Venn Diagrams: Probability as Area



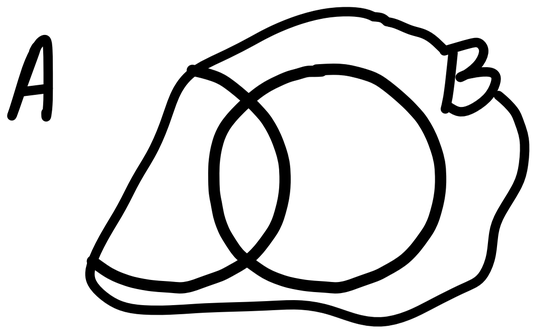
$P(A|B) = ?$

Venn Diagrams: Probability as Area



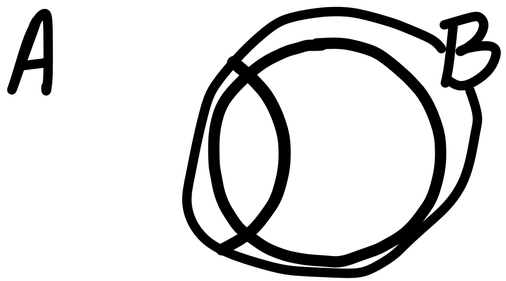
$P(A|B) = ?$

Venn Diagrams: Probability as Area



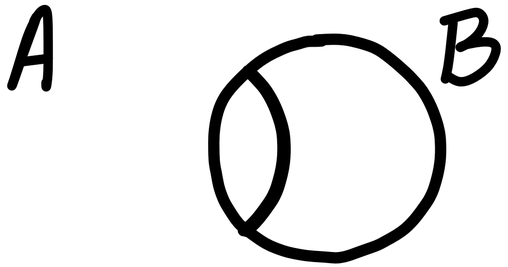
$P(A|B) = ?$

Venn Diagrams: Probability as Area



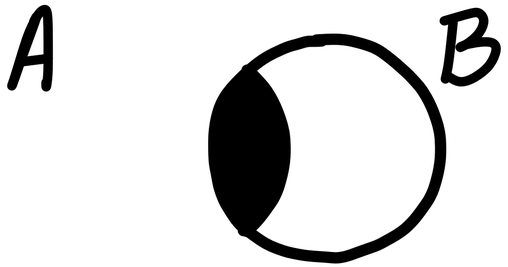
$P(A|B) = ?$

Venn Diagrams: Probability as Area



$P(A|B) = ?$

Venn Diagrams: Probability as Area



$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(\text{get a King} \mid \text{face card})$$

$$= \frac{P(\text{get a King } \underline{\underline{\text{and}}} \text{ face card})}{P(\text{face card})}$$

$$= \frac{P(\text{get a King})}{P(\text{face card})} = \frac{4/52}{12/52} = \frac{4}{12} = \boxed{\frac{1}{3}}$$

Conditional Probability With Random Variables

Let X and Y be discrete random variables.

We can talk about

$$P(X=x),\ P(Y=y),$$

and

$$P(X=x \text{ and } Y=y) \leftarrow \text{"joint probability"}$$

Joint Probabilities

$P(X = x \text{ and } Y = y)$

is written as

$$P(X = x, Y = y)$$

"and"

Joint Probabilities

- are often tabulated

		x		
		1	2	3
y	-1	0.20	0.10	0.10
	1	0.05	0.30	0.25

$P(X=2, Y=1)$

Joint Probabilities

- are often tabulated

		X		
		1	2	3
y	-1	0.20	0.10	0.10
	1	0.05	0.30	0.25

$P(X=2) = ?$

Joint Probabilities

- are often tabulated

		X		
		1	2	3
y	-1	0.20	0.10	0.10
	1	0.05	0.30	0.25

$$P(X=2) = 0.10 + 0.30 = 0.4$$

Joint Probabilities

- are often tabulated

		X			
		1	2	3	
y	-1	0.20	0.10	0.10	0.40
	1	0.05	0.30	0.25	0.60
		0.25	0.40	0.35	

$P(X=2)$ is a "marginal probability"

So, we have

$$P(X=x) = \sum_y P(X=x, Y=y)$$

$$P(Y=y) = \sum_x P(X=x, Y=y)$$

Let A = the event that X equals x.

Let B = the event that Y equals y.

Now $P(X=x, Y=y) = P(A \cap B)$,
and we have conditional
probabilities for random variables
already defined!

$$P(X=x \mid Y=y) = \frac{P(X=x, Y=y)}{P(Y=y)}$$